Raspberry Pi CCTV System (Motion + Google Drive)

📌 Project Goal

The objective of this project was to build a **reliable, low-cost CCTV system** using a Raspberry Pi that:

- Streams live video to a phone
- Records only when motion is detected
- Automatically uploads clips to Google Drive
- Does not fill up local storage
- Handles real-world issues like power loss

This system was implemented on **Raspberry Pi OS Bookworm** using Motion, with a focus on **event-driven design and reliability**.

🚀 Implementation Story

1️⃣ Preparing the system

First, the Raspberry Pi was updated.

```bash
sudo apt update
sudo apt upgrade -y
```

2️⃣ Installing required packages

The core components of the system are:

- Motion → video capture + motion detection
- FFmpeg → video encoding (used by Motion)
- rclone → cloud upload to Google Drive
- v4l-utils → camera debugging tools

All were installed using:

```
sudo apt install -y motion ffmpeg v4l-utils rclone
```

3 Verifying camera detection

Before configuring Motion, the camera was verified.

```
ls /dev/video*
v4l2-ctl --list-devices
```

Expected output:

```
/dev/video0
```

4 Configuring Motion

The main configuration file:

```
sudo nano /etc/motion/motion.conf
```

Key configuration:

```
daemon off
videodevice /dev/video0

stream_port 8081
stream_localhost off

webcontrol_port 8080
webcontrol_localhost off

target_dir /var/lib/motion

framerate 10

threshold 2000
noise_level 32
minimum_motion_frames 3

event_gap 2
pre_capture 3
```

```
post_capture 10

picture_output on
movie_output on
movie_codec mp4
movie_max_time 3

movie_filename %t-%v-%Y%m%d%H%M%S

on_movie_end /usr/local/bin/motion_upload.sh %f
```

5 Starting Motion

```
sudo systemctl enable motion
sudo systemctl restart motion
sudo systemctl status motion
```

6 Accessing live stream

Find Pi IP:

```
hostname -I
```

Open in browser:

http://<PI IP>:8081

7 Setting up Google Drive (rclone)

Initialize rclone:

```
rclone config
```

Create remote:

```
name: gdrive1
storage: drive
```

Verify:

```
rclone lsd gdrive1:
```

Create storage folder:

```
rclone mkdir gdrive1:CCTV_Motion_Clips
```

8 Making rclone accessible to Motion

Motion runs as a different user (`motion`), so its environment must be configured separately.

```
sudo mkdir -p /var/lib/motion/.config/rclone
sudo cp /home/aprem/.config/rclone/rclone.conf
/var/lib/motion/.config/rclone/rclone.conf
sudo chown -R motion:motion /var/lib/motion/.config
sudo chmod 700 /var/lib/motion/.config
sudo chmod 700 /var/lib/motion/.config/rclone
sudo chmod 600 /var/lib/motion/.config/rclone/rclone.conf
```

Verify access:

```
sudo -u motion rclone lsd gdrive1:
```

9 Creating upload script

```
sudo nano /usr/local/bin/motion_upload.sh
```

Script:

```
#!/bin/bash
set -euo pipefail

FILE="${1:-}"
DEST="gdrive1:CCTV_Motion_Clips"
LOG="/var/log/motion/rclone_motion.log"

[ -n "$FILE" ] || exit 0
[ -f "$FILE" ] || exit 0

prev_size=0
for i in {1..10}; do
```

```
  size=$(stat -c%s "$FILE" 2>/dev/null || echo 0)
  if [ "$size" -gt 0 ] && [ "$size" -eq "$prev_size" ]; then
    break
  fi
  prev_size="$size"
  sleep 0.3
done

/usr/bin/rclone copy "$FILE" "$DEST" \
  --log-file="$LOG" \
  --log-level INFO

rm -f "$FILE"
```

Make executable:

```
sudo chmod 755 /usr/local/bin/motion_upload.sh
sudo chown root:root /usr/local/bin/motion_upload.sh
```

10 Creating log directory

```
sudo mkdir -p /var/log/motion
sudo touch /var/log/motion/rclone_motion.log
sudo chown -R motion:motion /var/log/motion
sudo chmod 755 /var/log/motion
```

11 Restart Motion

```
sudo systemctl restart motion
```

12 Testing upload manually

```
sudo -u motion bash -lc 'echo test > /var/lib/motion/test.txt'
sudo -u motion /usr/local/bin/motion_upload.sh
/var/lib/motion/test.txt
```

Verify:

```
sudo -u motion rclone lsl gdrive1:CCTV_Motion_Clips
```

⚡ Power Loss Consideration

To reduce data loss:

```
movie_max_time 3
```

This ensures:

- frequent file closure
- frequent uploads
- max loss ≈ 3 seconds

📊 Final System Capabilities

- Live streaming via browser
- Motion-triggered recording
- Automatic Google Drive upload
- Minimal local storage usage
- Event-driven architecture
- Power-loss tolerant (within limits)
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- ## 🚧 Current Limitation

- No night vision (yet)
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- ## 📌 Useful Debug Commands

- ```
  sudo systemctl status motion
  sudo journalctl -u motion --no-pager -n 80
  sudo tail -n 80 /var/log/motion/motion.log
  sudo tail -n 80 /var/log/motion/rclone_motion.log
  ```
-
  ```

- ## 🗀 Important Paths

- Config → `/etc/motion/motion.conf`
- Videos → `/var/lib/motion`
- Script → `/usr/local/bin/motion_upload.sh`
- Logs → `/var/log/motion`
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- ## 🧠 Key Learnings

- Event-driven design > polling
- File stability matters before upload
- Service users require separate configs
- Real systems must handle failures
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